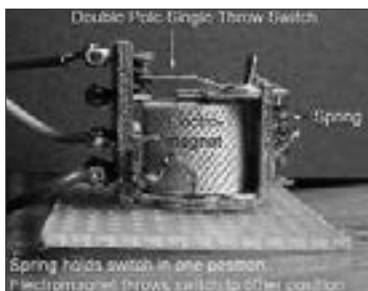


ry of computation. That paper solved a mathematical problem, but with a difference: the solution was derived not by mathematical calculations, but by *reasoning about the Turing machine*. The Turing Machine (TM) is a theoretical, simplified computer.



A basic relay. It is important, as you've gathered by now

That was it. Computer science was born. It should have correctly been called “computing science,” according to some, but well.

For this, we must quote stanford.edu to some length. Our explanations are in parentheses.

“Turing, writing before the invention of the modern digital computer, was interested in the question of what it means to be computable (meaning, simply, can it be computed? For example, the problem of factorising a 20,000-digit number is computable; the problem of determining c given $a + b + c = 20$ and $a = 3$ is not!). Intuitively a task is computable if one can specify a sequence of instructions which when followed



The face of computer science:
Alan Turing

will result in the completion of the task. Such a set of instructions is called an effective procedure, or algorithm, for the task. This intuition must be made precise by defining the capabilities of the device that is to carry out the instructions. Devices with different capabilities may be able to complete different instruction sets, and therefore may result in different classes of computable tasks.